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A STUDY ON BREAST CANCER PREDICTION BASED ON WDBC DATA

A DISSERTATION REPORT
SUBMITTED IN THE PARTIAL FULFILLMENT OF MASTER'S
DEGREE IN STATISTICS AND COMPUTING

UNDER THE SUPERVISION OF:

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CERTIFICATE

This is to certify that **Sayani Nag**, student of Master of Science in Statistics & Computing, semester IV of the Department of DST-CIMS, Institute of Science, Banaras Hindu University, bearing **exam roll no. 23419STC045** and **enrolment number 462955** of **2023-2025** has successfully completed her dissertation project entitled '**A study on breast cancer prediction based on WBDC data**'.

The project was completed under my direct supervision and guidance for the partial fulfilment of the course curriculum of Master of Science in Statistics & Computing. He carried out the whole work with utmost sincerity and the present report is the outcome of his sincere efforts and endeavours.

This work either in the present form or in part in any other form is not being submitted anywhere or for any other degree or diploma to the best of my knowledge.

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ABSTRACT

Breast cancer is one of the most prevalent cancers worldwide, necessitating early and accurate detection for effective treatment. This study explores the performance of machine learning classification techniques—**K-Nearest Neighbors (KNN)**, **Logistic Regression**, and **Support Vector Machine (SVM)**—in predicting breast cancer malignancy using the Wisconsin Diagnostic Breast Cancer dataset. The models were evaluated based on metrics such as **accuracy, precision, recall, and F1-score** to determine their effectiveness in distinguishing benign from malignant tumors. Preprocessing steps included **feature scaling, handling missing values, Outlier detection, multicollinearity check and data normalization** to enhance model performance. Results indicate that all three algorithms achieved high accuracy (>90%), with **SVM** being the best-performing model, achieving 97% accuracy, outperforming others. The study demonstrates the potential of machine learning in aiding clinical decision-making and highlights avenues for future work, such as hyperparameter tuning or ensemble methods for improved robustness.

INTRODUCTION

After lung cancer, the major reason for women's death is breast cancer. According to Indian Council for Medical Research, new cancer cases are to be 14.5 lakhs in 2016. Every year this number increased up to 17.3 lakhs in 2020. Different organizations play a crucial role for care of breast cancer in India to reduce the number of patients. Despite advancements in medical imaging, early and accurate diagnosis remains challenging due to the subjectivity of radiological assessments and the high false-positive rates of traditional screening methods. Machine learning (ML)-based classification techniques offer a data-driven, automated approach to improve diagnostic precision by detecting subtle patterns in clinical datasets.

This study evaluates and compares the performance of three supervised ML algorithms—**K-Nearest Neighbors (KNN)**, **Logistic Regression (LR)**, and **Support Vector Machine (SVM)**—in predicting breast cancer malignancy using the **Wisconsin Diagnostic Breast Cancer (WDBC) dataset**. The WDBC dataset provides quantitative features extracted from digitized fine-needle aspirate (FNA) images, making it ideal for training classification models.

- **Key objectives are:**

1. Assess model efficacy using metrics such as accuracy, precision, recall, and F1-score.
2. Identify the optimal classifier for distinguishing benign from malignant tumors.

The findings demonstrate that SVM outperforms KNN and LR, highlighting its robustness in handling high-dimensional medical data. This work contributes to the growing adoption of ML in oncology by providing empirical evidence for SVM's reliability in binary cancer classification. Future research could explore deep learning models or multi-modal data integration to further enhance predictive performance.

MATERIAL METHOD

❖ Data Source:

The Wisconsin Diagnostic Breast Cancer (WDBC) dataset is a widely used benchmark dataset in machine learning for breast cancer classification tasks. Here are the precise details about its source and key characteristics:

Source:

- Original Repository: UCI Machine Learning Repository
- Dataset Link: [Breast Cancer Wisconsin \(Diagnostic\) - UCI Machine Learning Repository](#)

Dataset Details:

★ Origin:

- Collected by Dr. William H. Wolberg at the University of Wisconsin Hospitals, Madison.
- Features are computed from digitized Fine Needle Aspirate (FNA) images of breast masses.

★ Features:

- 30 numeric features (mean, standard error, and worst values of 10 key cellular characteristics):
 - Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Concave Points, Symmetry, Fractal Dimension.
- Target Variable: Binary classification:
 - Malignant (M) vs. Benign (B).

★ Samples:

- Total: 569 instances (357 benign, 212 malignant).

.All the features name and the features description is given in the following table:

Table 1: Description of each and every feature on this dataset

Features Name	Features description
id	Unique id number for the patient sample
diagnosis	Diagnosis result M= Malignant B= Benign
radius_mean	Mean of distances from center to perimeter points
texture_mean	Mean of standard deviation of gray-scale values
perimeter_mean	Mean size of the perimeter of the tumor
area_mean	Mean area of the tumor
smoothness_mean	Mean of local variation in radius lengths
compactness_mean	Mean of $\text{perimeter}^2 / \text{area} - 1.0$
concavity_mean	Mean severity of concave portions of the contour
concave points_mean	Mean number of concave portions of the contour
symmetry_mean	Mean symmetry of the tumor shape
fractal_dimension_mean	Mean complexity of the contour (fractal dimension)
radius_se	Standard error of the radius
texture_se	Standard error of the texture

perimeter_se	Standard error of the perimeter
area_se	Standard error of the area
smoothness_se	Standard error of the smoothness
compactness_se	Standard error of the compactness
concavity_se	Standard error of the concavity
concave points_se	Standard error of the concave points
symmetry_se	Standard error of the symmetry
fractal_dimension_se	Standard error of the fractal dimension
radius_worst	Maximum value (worst) of radius
texture_worst	Maximum value (worst) of texture
perimeter_worst	Maximum value (worst) of perimeter
area_worst	Maximum value (worst) of area
smoothness_worst	Maximum value (worst) of smoothness
compactness_worst	Maximum value (worst) of compactness
concavity_worst	Maximum value (worst) of concavity
concave points_worst	Maximum value (worst) of concave points
symmetry_worst	Maximum value (worst) of symmetry
fractal_dimension_worst	Maximum value (worst) of fractal dimension

❖ Methodology

The analysis part of the data can be broadly classified into three part:

- **Data Preprocessing and EDA:** Performing data cleaning to handle missing values, outliers, duplicates in the dataset. Applying feature scaling using different methods like standardization or normalization on the numerical features for the uniformity. Encoding the categorical variables using the techniques like one hot encoding or label encoding. It also involves exploratory data analysis (EDA). EDA helps to explore and understand the structure, pattern and relationships within the dataset. It helps to gain insights from the dataset.
- **Building and evaluating the model:** For the model building part, the dataset has been split into two parts, 70% as training data and 30% as test data. Training data is used for construction of the model and test data is used for checking the validation of the model. Then selecting the appropriate algorithms based on the characteristics of the data and desired outcomes. For the Breast cancer disease prediction different models were used like:
 - Logistic Regression
 - K Nearest Neighbour (KNN)
 - Support vector machine (SVM)

Checking the performance of the models using different evaluation metrics. Common metrics for binary classification problems like Breast cancer disease prediction include accuracy, precision, recall, F1-score, Confusion matrix. Then compare the performance of different models to identify the most appropriate model.

Data Preprocessing and EDA:

➤ Dataset balance or unbalanced:

In machine learning, the balance of a dataset plays a crucial role in how well a model can learn and generalize. Let me explain the difference between balanced and imbalanced datasets in simple terms.

What is a Balanced Dataset?

A dataset is considered balanced when all classes or categories are represented equally, or nearly equally. This means each class has roughly the same number of samples, giving the model a fair chance to learn patterns from all categories without bias.

Example of a balanced dataset (binary classification):

- Class A: 500 samples
- Class B: 500 samples

Here, both classes have equal representation, so the model is not skewed toward predicting one over the other.

What is an Imbalanced Dataset?

An imbalanced dataset is when one or more classes dominate the dataset, while others are underrepresented. This imbalance can cause the model to favor the majority class, leading to poor performance on the minority class—even if overall accuracy seems high.

Example of an imbalanced dataset (binary classification):

- Class A: 950 samples
- Class B: 50 samples

In this case, Class A has **19 times more samples** than Class B. If we train a model on this data, it might simply learn to always predict Class A and still achieve **95% accuracy**, completely ignoring Class B.

Why Does This Matter?

- **Biased Predictions:** The model may perform well on the majority class but fail on the minority class.
- **Misleading Metrics:** Accuracy alone can be deceptive—you might need precision, recall, or F1-score to properly evaluate performance.
- **Real-World Impact:** In critical applications like medical diagnosis (e.g., cancer detection), missing the minority class (malignant cases) can have serious consequences.

How to Handle Imbalanced Data?

If your dataset is imbalanced, you can use techniques like:

1. **Resampling** (oversampling the minority class or undersampling the majority class).
2. **Synthetic Data Generation** (e.g., SMOTE).
3. **Class Weighting** (penalizing the model more for mistakes on the minority class).

Final Thought

We should always check our dataset's class distribution before training a model. A balanced dataset leads to fairer learning, while imbalanced data requires careful handling to avoid biased results.

➤ Missing value:

Missing values are defined as the values or data that is not stored (or not present) for some variable/s in the given dataset. Missing values are represented by Nan. It stands for Not a Number. There can be multiple reasons why certain values are missing from the data. Reasons for the missing data from the dataset affect the approach of handling missing data. So it's necessary to understand why the data could be missing.

Some of the reasons are listed below:

- Past data might get corrupted due to improper maintenance.

- Observations are not recorded for certain fields due to some reasons. There might be a failure in recording the values due to human error.
- The user has not provided the values intentionally
- Item nonresponse: This means the participant refused to respond.

Handling Missing Values:

Deletion: This involves removing rows or columns with missing values.

Imputation: This replaces missing values with estimates.

- ★ **Mean/Median/Mode Imputation:** Replace missing entries with the average (mean), middle value (median), or most frequent value (mode) of the corresponding column.
- ★ **K-Nearest Neighbors (KNN Imputation):** This method finds the closest data points (neighbors) based on available features and uses their values to estimate the missing value.
- ★ **Model-based Imputation:** This involves creating a statistical model to predict the missing values based on other features in the data.

➤ **Outliers**

An outlier refers to a data point that significantly deviates from the rest of the observations. It can occur due to various reasons such as measurement variability, the presence of new or unusual data, or experimental errors, which are sometimes excluded from the dataset. Outliers can provide valuable insights or indicate interesting patterns in the data, but they can also pose challenges when conducting statistical analyses.

Outliers detection and removal:

- ★ **Trimming-** This technique removes any outlier values from our analysis, resulting in a narrower dataset when there are more outliers present. Its primary benefit is its speediness.

- ★ **Capping-** In this technique called “outlier detection,” we cap our data to set limits. For instance, if we decide on a specific value, any data point above or below that value is considered an outlier. The number of outliers in the dataset then gives us insight into that capping number. It’s like setting a boundary.
- ★ **Discretization-** In the method of outlier detection, we create groups and categorize the outliers into a specific group, making them follow the same behavior as the other points in that group. This approach is often referred to as Binning. Binning is a way of organizing data, especially in outlier detection, where we group similar items together, helping us identify and understand patterns more effectively.

For Normal Distributions

- Use empirical relations of Normal distribution.
- The data points that fall below $\text{mean} - 3 \times (\text{sigma})$ or above $\text{mean} + 3 \times (\text{sigma})$ are outliers, where mean and sigma are the average value and standard deviation of a particular column.

For Skewed Distributions

- Use **Inter-Quartile Range (IQR)** proximity rule.
- The data points that fall below **first quartile (Q1) – 1.5 IQR** or above the **third quartile $Q3 + 1.5 \text{ IQR}$** are outliers, where Q1 and Q3 are the 25th and 75th percentile of the dataset, respectively. IQR represents the inter-quartile range and is given by $Q3 - Q1$.

For Other Distributions

- Use a percentile-based approach. For example, data points that are far from the 99% percentile and less than 1 percentile are considered an outlier

Box plot method: A box plot visually represents the distribution of data and identifies outliers based on the position of data points outside the whiskers (the lines extending from the box). Outliers can be identified and removed based on certain criteria, such as being above or below a specific multiple of the interquartile range.

➤ Correlation:

Correlation refers to the degree to which two or more variables are related or move together. It quantifies the strength and direction of the linear relationship between variables. Correlation coefficient is often denoted by the symbol "r" which ranges between -1 to +1.

A positive correlation ($0 < r < 1$) indicates that as one variable increases, the other variable also tends to increase. The variables move in the same direction.

A negative correlation ($-1 < r < 0$) indicates that as one variable increases, the other variable tends to decrease. The variables move in opposite directions.

A correlation coefficient close to 0 ($r \approx 0$) indicates a weak or no linear relationship between the variables.

The magnitude of the correlation coefficient indicates the strength of the relationship. The closer the correlation coefficient is to -1 or +1, the stronger the relationship.

We have used different techniques to visualize correlation

Pairplot: Pairplots are useful for exploring relationships between multiple variables in a dataset, identifying patterns, and detecting potential correlations.

Correlation heatmap: Heatmaps are useful for visualizing correlation matrices, where the intensity of colors indicates the strength and direction of correlations between variables.

➤ Features Scaling:

Feature Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing to handle highly varying magnitudes or values or units. If feature scaling is not done, then a machine learning algorithm tends to weigh greater values, higher and consider smaller values as the lower values, regardless of the unit given below:

Absolute Maximum Scaling

This method of scaling requires two-step:

1. We should first select the maximum absolute value out of all the entries of a particular measure.
2. Then after this, we divide each entry of the column by this maximum value.

$$X_{scaled} = \frac{X_i - \max(|X|)}{\max(|X|)}$$

After performing the above-mentioned two steps we will observe that each entry of the column lies in the range of -1 to 1.

Min-Max Scaling

This method of scaling requires below two-step:

1. First, we are supposed to find the minimum and the maximum value of the column.
2. Then we will subtract the minimum value from the entry and divide the result by the difference between the maximum and the minimum value.

$$X_{scaled} = \frac{X_i - X_{min}}{X_{max} - X_{min}}$$

Normalization

We subtract each entry by the mean value of the whole data and then divide the results by the difference between the minimum and the maximum value.

$$X_{scaled} = \frac{X_i - X_{mean}}{X_{max} - X_{min}}$$

Standardization

1. First, we should calculate the mean and standard deviation of the data we would like to normalize.
2. Then we are supposed to subtract the mean value from each entry and then divide the result by the standard deviation. This helps us achieve a normal distribution(if it is already normal but skewed) of the data with a mean equal to zero and a standard deviation equal to 1.

$$X_{scaled} = \frac{X_i - X_{mean}}{\sigma}$$

In the project we have used the method of standardization for scaling the numeric variables.

Label Encoding

It is the process of converting **categorical text data** (like "Apple", "Banana", "Cherry") into **numeric labels** (like 0, 1, 2). Each unique category gets assigned a different integer.

Key points:

- Used to make text data numerical for machine learning models.
- Introduces an *artificial order* (which can sometimes be a problem if categories don't actually have order).

Model building and Evaluation

Train Test Split: It is the process of splitting a dataset into two separate sets: one for training a model and the other for testing its performance.

- **Training Set:** This subset is used to train the machine learning model. It typically contains the majority of the data, allowing the model to learn patterns and relationships between features and labels.
- **Test Set:** This subset is used to evaluate the performance of the trained model.

❖ **Model used:**

❖ **Logistic Regression:**

Logistic regression is the appropriate regression analysis to conduct when the **dependent variable is dichotomous (binary)**. Like all regression analyses, logistic regression is a predictive analysis. It is used to describe data and to explain the relationship between one dependent binary variable and one or more nominal, ordinal, interval or ratio-level independent variables. **Logistic regression uses the logistic function (also called the sigmoid function) to map the output of the linear combination of features to a probability value between 0 and 1.**

$$P(y=1|x) = \frac{1}{1+e^{(b_0+b_1x_1+\dots+b_nx_n)}}$$

Where;

- $P(y=1|x)$ is the probability of the target variable being 1 given input x .
- e is the base of the natural logarithm.
- $b_0, b_1, \dots, b_n$ are the coefficients of the logistic regression model.
- $x_1, \dots, x_n$ are the independent variables (features).

The logistic regression model is trained using optimization techniques such as maximum likelihood estimation (MLE) or gradient descent. The goal is to find the optimal coefficients that maximize the likelihood of the observed data. In binary classification, logistic regression uses a decision boundary to classify the input into one of the two classes based on the predicted probability. By default, the decision boundary is set at 0.5, but it can be adjusted depending on the specific requirements of the problem.

Common evaluation metrics for logistic regression include accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve (AUC-ROC).

Accuracy: It represents the proportion of correctly classified instances out of the total instances in the dataset.

Recall: It evaluates a model's ability to correctly identify all positive instances (i.e., true positives) out of all actual positive instances in the dataset.

F1 Score: It combines both precision and recall into a single metric, providing a balanced assessment of the model's accuracy.

Confusion matrix: It allows visualization of the performance of an algorithm.

❖ K Nearest Neighbour (KNN):

KNN is a straightforward supervised learning algorithm used for:

- Classification (e.g., "Is this tumor malignant?")
- Regression (e.g., "Predicting house prices")

How It Works

1. **Memorize the Data:** Unlike other algorithms, KNN doesn't "learn" during training—it just stores the dataset (*lazy learner*).
2. **Predict New Points:** For a new data point:
 - Calculate distances (e.g., Euclidean distance) to all stored points.
 - Pick the 'k' closest neighbors (e.g., **k=5**).

- **Classification:** Majority vote decides the class.
- **Regression:** Average of neighbors' values.

Key Characteristics

- No Training Phase: Works directly with data (fast training, slower predictions).
- Hyperparameter 'k': Small **k** = sensitive to noise; large **k** = smoother but less precise.
- Feature Scaling Matters: Distance-based, so normalize features (e.g., use **StandardScaler**).
- Curse of Dimensionality: Struggles with high-dimensional data (many features).

When to Use?

- Small to medium datasets.
- Simple baseline model.
- Interpretability matters (you can see why a prediction was made).

Example:

If **k=3** and 2 neighbors are "Malignant" while 1 is "Benign," the prediction is "Malignant."

Common evaluation metrics for KNN include accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve (AUC-ROC).

Accuracy: It represents the proportion of correctly classified instances out of the total instances in the dataset.

Recall: It evaluates a model's ability to correctly identify all positive instances (i.e., true positives) out of all actual positive instances in the dataset.

F1 Score: It combines both precision and recall into a single metric, providing a balanced assessment of the model's accuracy.

Confusion matrix: It allows visualization of the performance of an algorithm

❖ SVM

SVM is a supervised learning model for classification (and regression) that finds the optimal hyperplane separating classes with the maximum margin.

Key Concepts

1. Hyperplane

- A decision boundary (e.g., a line in 2D, plane in 3D) that separates classes.
- Defined by: $W^T x + b = 0$
- where:
 - w = weight vector (normal to the hyperplane),
 - b = bias term.

2. Support Vectors

- The closest data points to the hyperplane (they "support" the margin).

3. Margin

- The distance between the hyperplane and the nearest data points of either class.
- SVM maximizes this margin for better generalization.

Loss Function: Hinge Loss

SVM uses the hinge loss to enforce margin maximization and classification correctness:

$$L(W, b) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \max(0, 1 - y_i(w^T x_i + b))$$

$\frac{1}{2} \|w\|^2$ denotes L2 regularization (penalizes large weights to avoid overfitting).

$\max(0, 1 - y_i(w^T x_i + b))$ denotes ,

- Hinge loss

- y_i = true label
- $w^T x_i + b$ = predicted score

How Hinge loss works?

- If $w^T x_i + b \geq 1$ then Point is correctly classified outside the margin
- If $w^T x_i + b < 1$ point is inside the margin or misclassified

Parameter C

- Controls the trade-off between margin width and misclassification penalty.
 - **Small C:** Prioritizes wider margin (may tolerate some misclassifications).
 - **Large C:** Strict classification (narrower margin, fewer misclassifications).

Common evaluation metrics for SVM include accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve (AUC-ROC).

Accuracy: It represents the proportion of correctly classified instances out of the total instances in the dataset.

Recall: It evaluates a model's ability to correctly identify all positive instances (i.e., true positives) out of all actual positive instances in the dataset.

F1 Score: It combines both precision and recall into a single metric, providing a balanced assessment of the model's accuracy.

Confusion matrix: It allows visualization of the performance of an algorithm.

Result

Results of diagram:

Fig 1: This is a pie chart showing the distribution of variables called **Diagnosis**.

The two categories are:
B (Benign) — 62.7%
M (Malignant) — 37.3%

This diagram as well as explained that our data is balanced

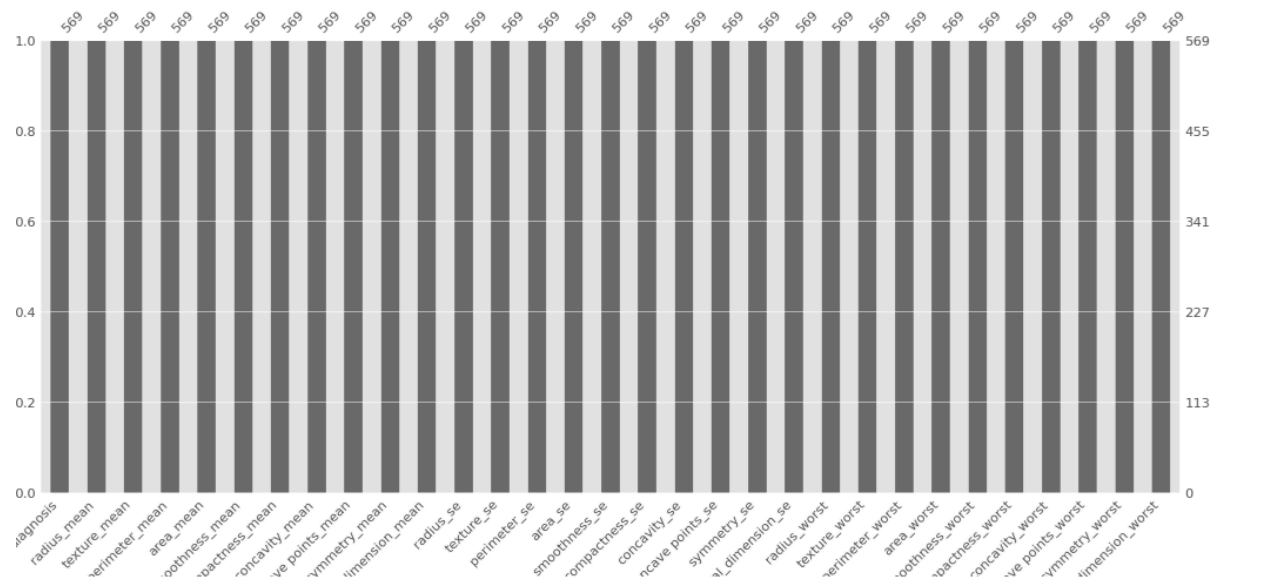
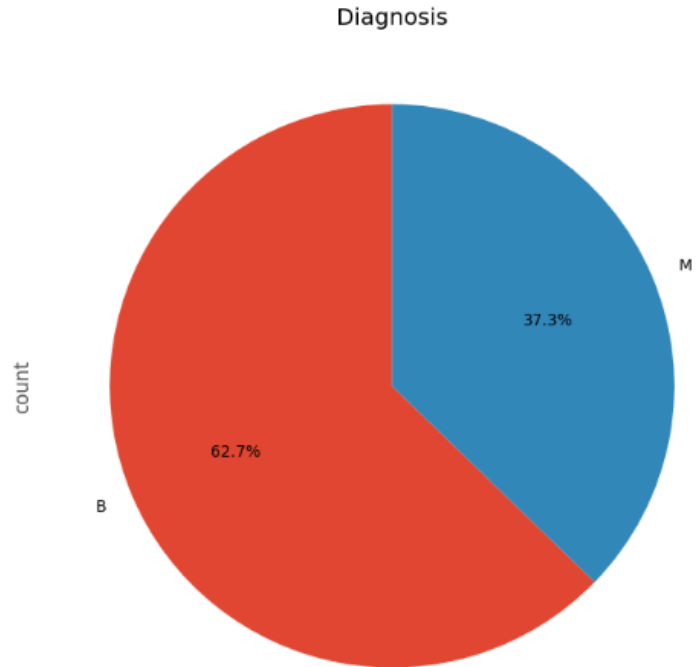


Fig 2:MSNO BAR .This is the visualization that **there is no missing value present**

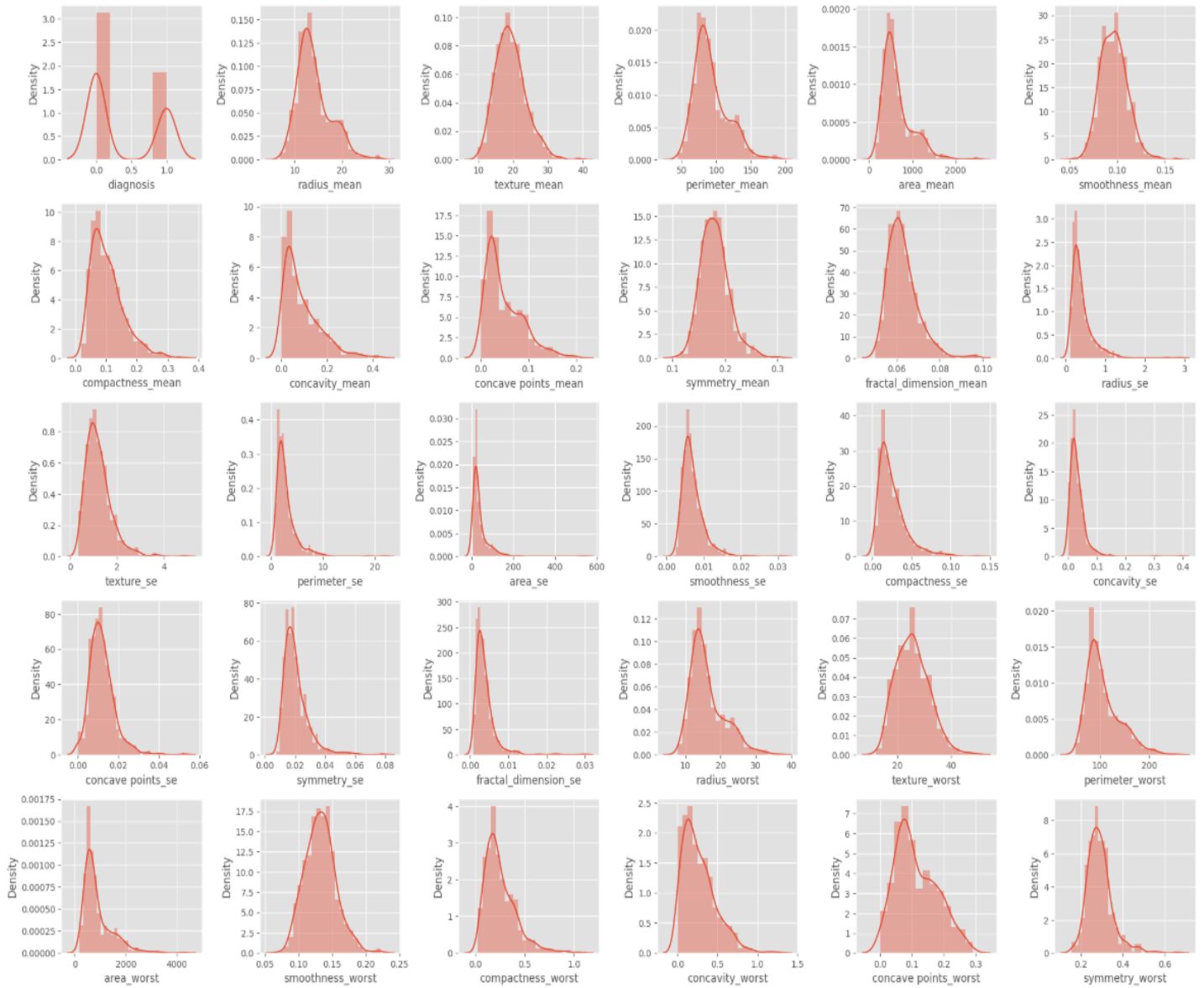


Fig 3: The underlying distributions of each and every feature. From the above diagram we can state that :

- the diagnosis column has two different peak and its shows binary distribution
- some columns are like area mean concave points mean has right skewed distribution
- some columns are like texture mean and symmetry mean has symmetric distribution

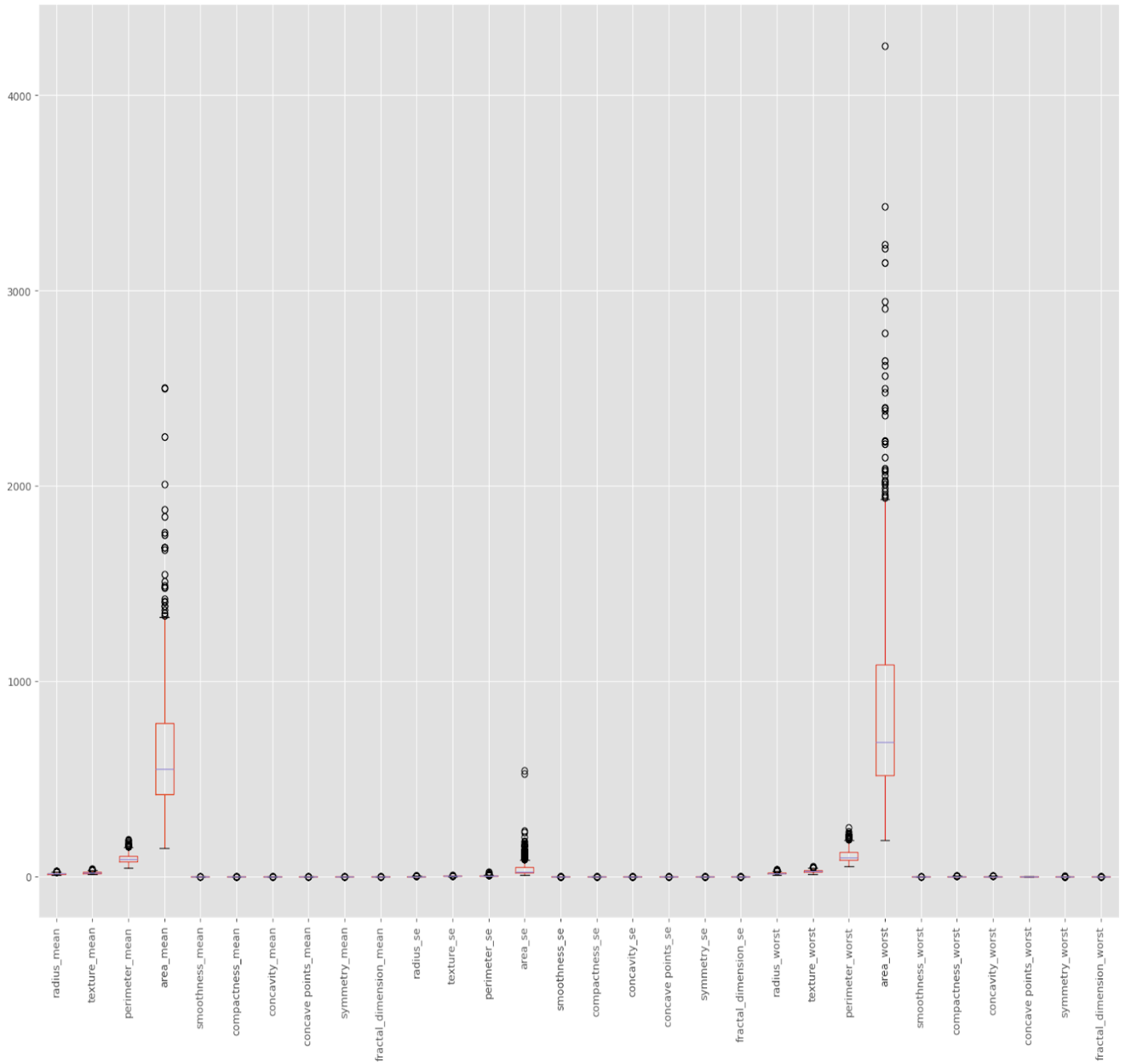


Fig 4: Box plot of every feature. This image shows that the feature like area mean perimetric mean and area worst has outliers.

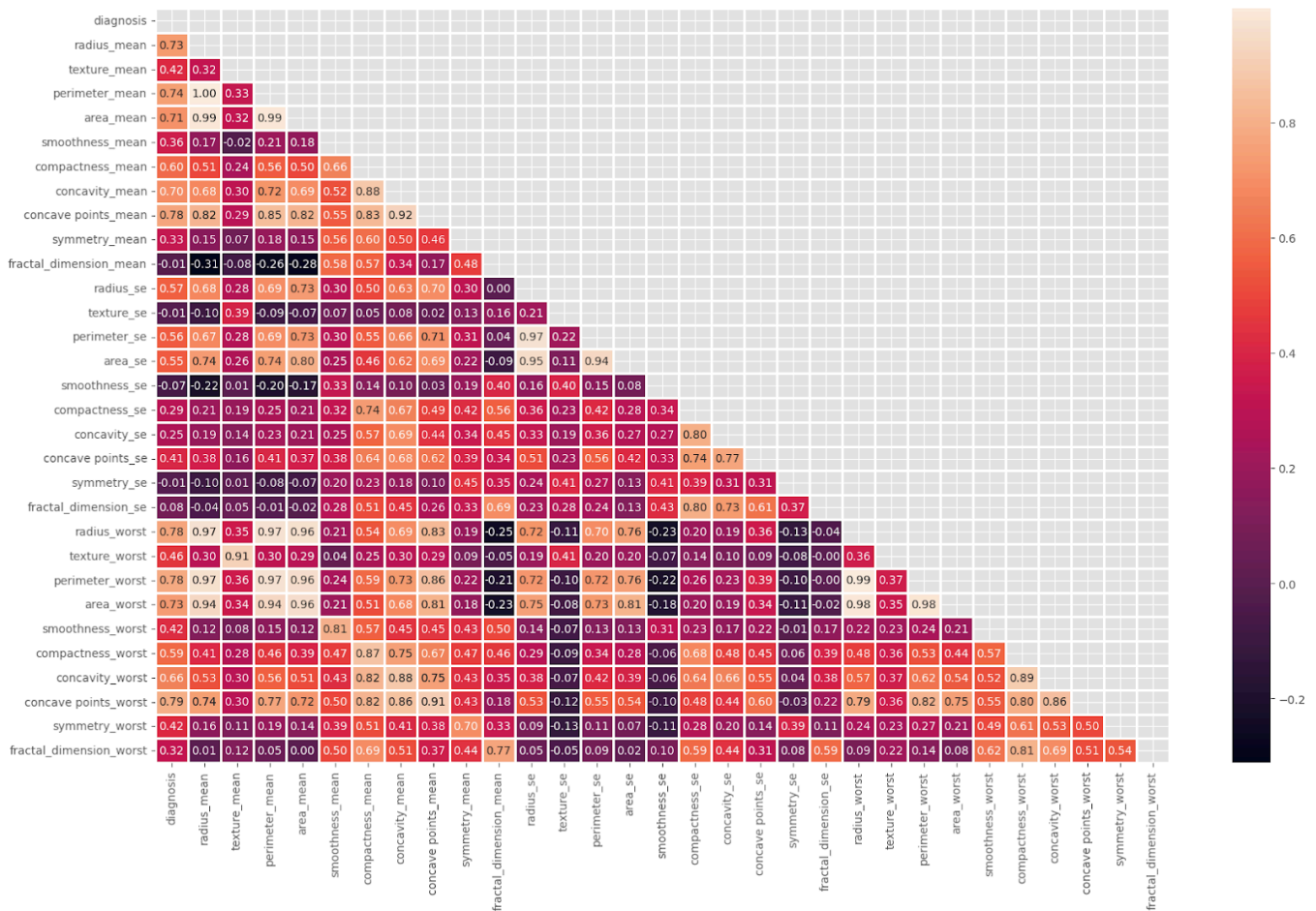


Fig 5: Heatmap. This explains that:

- feature related to the size of the cell of nucleus(radius mean,perimeter mean area mean,radius worst , perimeter worst, area worst) are strongly correlated with each other
- features like compactness mean smoothness mean concavity mean and concave summeans has moderate to higher positive correlation
- the feature diagnosis has moderate positive correlation with several mean measurement which implies that the larger cell size associated with a diagnosis of malignant tumors

Result of Model performance

❖ Logistic Regression:

Table-2 :Confusion matrix of logistic Regression

Actual	Predicted	
	0	1
0	105	2
1	4	60

True positive(TP) = 60

True negative(TN) = 105

False positive(FP) = 2

False negative(FN) = 4

Model accuracy is 96%

❖ KNN:

Table-3 :Confusion matrix of KNN

Actual	Predicted	
	0	1
0	102	5
1	7	57

True positive(TP) = 57

True negative(TN) = 102

False positive(FP) = 5

False negative(FN) = 7

Model accuracy is 93%

❖ SVM

Table-3 :Confusion matrix of KNN

Actual	Predicted	
	0	1
0	105	2
1	3	61

True positive(TP) = 61

True negative(TN) = 105

False positive(FP) = 2

False negative(FN) = 3

Model accuracy is 97%

As we know our dataset is balanced this implies that we can draw conclusions about model performance based on the accuracy whereas using SVM we can achieve highest accuracy.

Conclusion

In this project, we proposed three methods in which comparative analysis was done and results were achieved. The conclusion which we found is that machine learning algorithms performed better in this analysis. The methods used to compare the model performance is the Confusion matrix. Also as from the pie chart we get that our dataset is balanced so based on the accuracy we can easily evaluated which ml algorithm works better in this dataset

For the 32 features which were in the dataset, Support vector classifier(SVM) performed better in the ML approach when data preprocessing is applied which includes detecting the presence of missing value , presence of outlier multicollinearity.

We got an accuracy of 96% in logistic regression. In the case of KNN it is found that the number of false positives and the number of false negatives is relatively higher than the other two models that are used in this project . But we still get an accuracy of 93% in KNN. Then applied SVM which gives us the highest accuracy of 97% . In the case of SVM we used the RBF kernel which also helps us to get an accurate margin . Considering the accuracy of each model we can say that SVM works better in the dataset.

Discussion

Previous Work on the Wisconsin Diagnostic Breast Cancer (WDBC) Dataset.

The WDBC dataset has been extensively studied in machine learning and medical research. Below is a summary of key approaches, models, and findings from prior work:

★ . Popular Machine Learning Models Applied

Researchers have experimented with various algorithms to classify tumors as **benign (B)** or **malignant (M)** using WDBC:

1. Traditional ML Models

Support Vector Machines (SVM): Achieved 95–98% accuracy - Kernels like RBF and linear were commonly used.

Logistic Regression: Reported 92–96% accuracy, often with L2 regularization.

k-Nearest Neighbors (k-NN): Performance varied (90–97%) accuracy based on k-value and distance metric (Euclidean/Manhattan).

Decision Trees & Random Forest: Random Forest often outperform single trees (96% accuracy). Feature importance analysis (e.g., worst radius, concave points were top predictors).

Neural Networks (NLP): Early studies used shallow networks (94–97% accuracy).

2. Ensemble & Hybrid Models

AdaBoost/XGBoost improved accuracy (97–98%) by combining weak learners.

SVM + PCA: Dimensionality reduction (PCA) sometimes boosted performance.

★ Key Findings from Literature

1. Feature Importance

Top predictive feature: `worst radius` , `worst concave points` `mean perimeter`

Less useful: `fractal dimension` or `texture` in some studies.

2. Data Preprocessing Matters

Standardization (e.g., `StandardScaler`) was critical for distance-based models (SVM, k-NN).

Outlier removal improved robustness in some cases.

3. Class Imbalance Handling:

WDBC is moderately imbalanced (62% benign, 38% malignant).

Some studies used **SMOTE** or **class weighting** to improve recall for malignant cases.

4. Performance Metrics Beyond Accuracy:

Precision, Recall (Sensitivity), F1-score were emphasized due to medical implications. Example: A model with 98% accuracy but 85% recall for malignant cases was considered suboptimal.

★ Limitations & Challenges Identified

Curse of Dimensionality: With 30 features, some models (like k-NN) suffered unless PCA was applied.

Generalization Issues: High accuracy on WDBC didn't always translate to real-world clinical data.

Interpretability Trade-offs: Black-box models (e.g., neural networks) lacked explainability for medical use.

★ How myWork Fits In

My results (97% with SVM) align with top-performing traditional ML models. Potential contributions:

Comparative analysis of SVM vs. other models on WDBC.

Hyperparameter tuning insights (e.g., optimal `C` and kernel for SVM).
Explainability (e.g., visualizing support vectors or decision boundaries).

★ Key Papers/Datasets for Citation

1. Original WDBC paper:

Street, W.N., Wolberg, W.H., & Mangasarian, O.L. (1993). Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE's Symposium on Electronic Imaging.

2. Benchmark studies:

- Dua, D. & Graff, C. (2019). UCI Machine Learning Repository.
- Aljarah et al. (2018). Optimizing SVM parameters for WDBC classification.*

Reference

❖ *Original WDBC Dataset & Early Studies*

- **Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993).**
 - Nuclear feature extraction for breast tumor diagnosis.
 - IS&T/SPIE's Symposium on Electronic Imaging.
 - [DOI:10.1117/12.148698](https://doi.org/10.1117/12.148698)
 - This is the original study introducing the WDBC dataset.
- **Dua, D., & Graff, C. (2019).**
 - UCI Machine Learning Repository.
 - [WDBC Dataset Link](#)
 - Official repository hosting the dataset with benchmarks.

❖ *Machine Learning on WDBC (SVM, k-NN, RF, etc.)*

- **Mangasarian, O. L., & Wolberg, W. H. (1990).**
 - Cancer diagnosis via linear programming.
 - SIAM News, 23(5), 1–18.
 - Early work using linear programming (pre-SVM) on WDBC.
- **Çomak, E., et al. (2007).**
 - A decision support system based on SVM for diagnosis of breast cancer.
 - Expert Systems with Applications, 32(4), 939–946.
 - [DOI:10.1016/j.eswa.2006.01.022](https://doi.org/10.1016/j.eswa.2006.01.022)
 - SVM-based classification on WDBC.
- **Karabatak, M., & Ince, M. C. (2009).**
 - An expert system for detection of breast cancer based on association rules and neural network.
 - Expert Systems with Applications, 36(2), 3465–3469.
 - Hybrid approach (association rules + NN) on WDBC.

❖ *Feature Importance & Preprocessing*

- **Guyon, I., et al. (2002).**
 - Gene selection for cancer classification using support vector machines.

- **Machine Learning*, 46(1-3), 389–422.*
- [DOI:10.1023/A:1012487302797](https://doi.org/10.1023/A:1012487302797)
- Discusses feature selection methods (relevant to WDBC's 30 features).
- **Liu, H., et al. (2005).**
 - *Feature selection for high-dimensional data: A fast correlation-based filter solution.*
 - *ICML Workshop on Feature Selection.*
 - *Methods applicable to WDBC's feature space*
- ❖ **Class Imbalance & Performance Metrics**
- **Chawla, N. V., et al. (2002).**
 - *SMOTE: Synthetic Minority Over-sampling Technique.*
 - *JAIR*, 16, 321–357.
 - [DOI:10.1613/jair.953](https://doi.org/10.1613/jair.953)
 - *Seminal paper on handling imbalance (relevant to WDBC's 62:38 split).*
- **Sokolova, M., & Lapalme, G. (2009).**
 - *A systematic analysis of performance measures for classification tasks.*
 - *Information Processing & Management*, 45(4), 427–437.
 - *Explains why accuracy alone is insufficient for WDBC.*